SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5115
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I __________Meghanadh A_(192372281)________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

2 A)

Block Cipher Mode Evaluation

Given:

- Plaintext Blocks:
  - P_1 = 1010
  - P_2 = 1010
  - P_3 = 1111
  - P_4 = 1010
- **Key (K) = 1100**
- **IV = 0000** (for CBC)

Assumption:
Encryption is performed using the XOR operation:

$C=P\oplus KC$ = P \oplus KC=$P\oplus K$

(a) Encrypt using ECB and CBC Modes

1. ECB (Electronic Codebook)

Formula:

$Ci=Pi\oplus K$

Plaintext	Key	Ciphertext
1010	1100	0110
1010	1100	0110
1111	1100	0011
1010	1100	0110

ECB Ciphertext:

C = [0110, 0110, 0011, 0110]

2. CBC (Cipher Block Chaining)

Formula:

Ci=(Pi⊕Ci−1)⊕KC_i=(P_i\oplus C_{i-1})\oplus KCi=(Pi⊕Ci−1)⊕K

Where:

- IV = 0000
- C0=IVC_0 = IVC0=IV

Block 1

P_1 = 1010

$P_1 \oplus$ IV

1010 ⊕ 0000 = 1010

Encrypt:

1010 ⊕ 1100 = **0110**

C_1 = 0110

Block 2

P_2 = 1010

$P_2 \oplus C_1$

1010 ⊕ 0110 = 1100

Encrypt

1100 ⊕ 1100 = **0000**

C_2 = 0000

Block 3

P_3 = 1111

1111 ⊕ 0000 = 1111

1111 ⊕ 1100 = **0011**

C_3 = 0011

Block 4

P_4 = 1010

1010 ⊕ 0011 = 1001

1001 ⊕ 1100 = **0101**

C_4 = 0101

CBC Ciphertext:

C = [0110, 0000, 0011, 0101]

(b) Compare the Ciphertext Outputs

Block	ECB	CBC
C_1	0110	0110
C_2	0110	0000
C_3	0011	0011
C_4	0110	0101

Observation:

- In **ECB**, identical plaintext blocks (`1010`) produce the same ciphertext (`0110`).
- In **CBC**, the repeated plaintext blocks generate different ciphertexts because each block depends on the previous ciphertext.

(c) Which Mode Hides Patterns Better?

CBC mode hides patterns better than ECB mode.

Explanation:

- **ECB Mode:** Each plaintext block is encrypted independently. Therefore, identical plaintext blocks produce identical ciphertext blocks, making patterns visible and reducing security.
- **CBC Mode:** Before encryption, each plaintext block is XORed with the previous ciphertext block (or the IV for the first block). This causes identical plaintext blocks to produce different ciphertexts when their preceding ciphertexts differ, effectively hiding patterns and providing stronger security.

Conclusion

- **ECB Ciphertext: [0110, 0110, 0011, 0110]**
- **CBC Ciphertext: [0110, 0000, 0011, 0101]**
- **Better Mode: CBC**, because it conceals repeated plaintext patterns by chaining each block with the previous ciphertext block before encryption.

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

3A)

DES vs 3-DES vs Blowfish Performance Study

Given:

- **DES**
  - Key Size = **56 bits**
  - Encryption Time = **2 ms/block**
- **3-DES**
  - Key Size = **168 bits**
  - Encryption Time = **6 ms/block**
- **Blowfish**
  - Key Size = **128 bits**
  - Encryption Time = **1 ms/block**
- **Number of Data Blocks = 1000**

(i) Calculate Total Encryption Time
Formula:
Total Encryption Time=Time per Block×Number of Blocks

Algorithm	Time per Block	Blocks	Total Time
DES	2 ms	1000	**2000 ms = 2 seconds**
3-DES	6 ms	1000	**6000 ms = 6 seconds**
Blowfish	1 ms	1000	**1000 ms = 1 second**

- DES = 2000 ms (2 s)
- 3-DES = 6000 ms (6 s)
- Blowfish = 1000 ms (1 s)

(ii) Analyze Trade-offs Between Security and Performance

Algorithm	Key Size	Security	Performance	Remarks
DES	56 bits	Low	Fast	Small key size makes it vulnerable to brute-force attacks.
3-DES	168 bits	High	Slow	Very secure but requires three DES operations, making it slower.
Blowfish	128 bits	High	Very Fast	Provides strong security with faster encryption than DES and 3-DES.

Analysis:

- **DES** has the fastest performance after Blowfish but offers the **least security** due to its 56-bit key.

- **3-DES** provides **strong security** but is **three times slower** than DES.
- **Blowfish** offers a **good balance** between security and speed, making it efficient for encrypting large amounts of data.

(iii) Recommend the Best Algorithm for Secure Real-Time Communication

Recommended Algorithm: Blowfish

Reasons:

- Uses a **128-bit key**, providing strong security.
- Has the **fastest encryption time (1 second for 1000 blocks)**.
- Suitable for **real-time communication** because of its high speed and strong protection against attacks.
- More efficient than 3-DES while maintaining a high level of security.

Algorithm	Total Time	Security	Suitable for Real-Time?
DES	2s	Low	No
3-DES	6s	High	No (too slow)
Blowfish	1s	High	Yes (Recommended)

RUBRICS FOR CALCULATION:

Criteria	Weighta ge	Level 4 (Exemplary)	Level 3 (Proficie nt)	Level 2 (Developi ng)	Level 1 (Begin nin g)
Problem Understa ndi ng	20%	Clearly interprets problem, identifies all parameter s and requiremen ts accurately	Understa n ds problem with minor gaps	Partial understa ndi ng; misses key elements	Misinter pre ts or unable to understa nd problem
Method Selection	20%	Selects most appropriate method/for m ula with	Selects correct method with minor errors	Inappropri at e or partially correct method	Unable to select method

STRONG JUSTIFICATION

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorgani ze d steps; multiple errors	No clear process
Accuracy of Calculatio n	20%	All calculatio ns accurate; correct final answer	Minor calculati on errors	Frequent errors; incorrect final answer	No valid calculati on s
Interpreta tio n of Results	15%	Clearly interprets results with units and engineeri ng relevance	Basic interpret ati on with minor omissions	Limited or unclear interpreta tio n	No interpret ati on